Multivariate Random Signal Characterization

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Key Concepts

1) A random vector is a collection of scalar random variables, x_i for $i = 1, 2, \ldots N$.

2) The probability density function of a random vector completely characterizes all the relationships between all random variables in the random vector.

3) The covariance matrix is the expected value (average over) the outer product of the difference between the random vector and the mean with itself.

4) The covariance matrix describes the relationships between all pairs of random variables.

    a) The diagonal gives the variance of the corresponding element of the random vector.

    b) Gaussian is the familiar bell-shaped probability density function. Described completely by its mean and standard deviation.

    c) The i, j element of the covariance matrix, r_{ij} , indicates how the ith and jth elements co vary.

        i) If $r_{ij} = 0$, then on average the fluctuations in x_i and x_j are unrelated. We say that x_i and x_j are uncorrelated.

ii) If $r_{ij} > 0$, then on average x_i and x_j tend to rise/fall together. When x_i is greater than its mean, x_j tends to be greater than its mean also.

iii) If $r_{ij} < 0$, then on average x_i and x_j tend to move in opposite directions. When x_i is greater than its mean, x_j tends to be less than its mean.

5) The correlation coefficient is the cross-correlation r_{ij} normalized by the standard deviation of x_i and x_j .

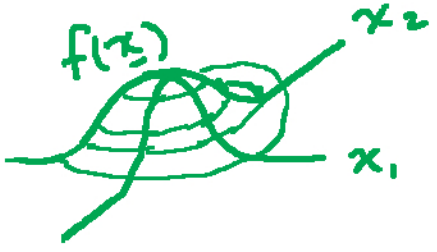
6) The covariance matrix for a multivariate Gaussian random vector defines the geometry of the probability density function.

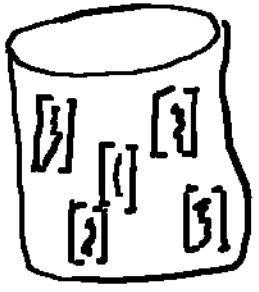
Video Screenshots

Vector: collection of scalars $x_1, x_2, \cdots x_N$

$\underline{x} = [x_1 \; x_2 \cdots x_N]^T$ $(N \times 1)$ $[\cdot]^T$ matrix transpose

pdf $f(\underline{x})$: characterizes each x_i and the relationships between all x_i's





Example: Multivariate Gaussian

$$f(\underline{x}) = (2\pi)^{-N/2} (\det \underline{R})^{-1/2} \exp\left\{-\frac{1}{2}(\underline{x}-\underline{m})^T \underline{R}^{-1} (\underline{x}-\underline{m})\right\}$$

$\underline{m}$: $(N \times 1)$ mean $\qquad (\cdot)^{-1}$ matrix inverse

$\underline{R}$: $(N \times N)$ covariance matrix

$\underline{m} = E\{\underline{x}\}$ so $[\underline{m}]_i = E\{x_i\}$, $\quad \underline{R} = E\{(\underline{x}-\underline{m})(\underline{x}-\underline{m})^T\} =$

Covariance matrix describes "link" between x_i and x_j

$r_{ij} = [\underline{R}]_{ij} = E\{(x_i - m_i)(x_j - m_j)\}$ how x_i and x_j co-vary

1) $i = j$ $r_{ii} = E\{(x_i - m_i)^2\}$ variance of x_i (diagonal of $\underline{R}$)

2) $i \neq j$ $r_{ij} > 0$ "on average" $x_i - m_i$ and $x_j - m_j$ have same sign $\Rightarrow$ x_i and x_j tend to rise/fall together

$r_{ij} < 0$ "on average" $x_i - m_i$ and $x_j - m_j$ have the opposite sign $\Rightarrow$ x_i and x_j tend to move in opposite directions

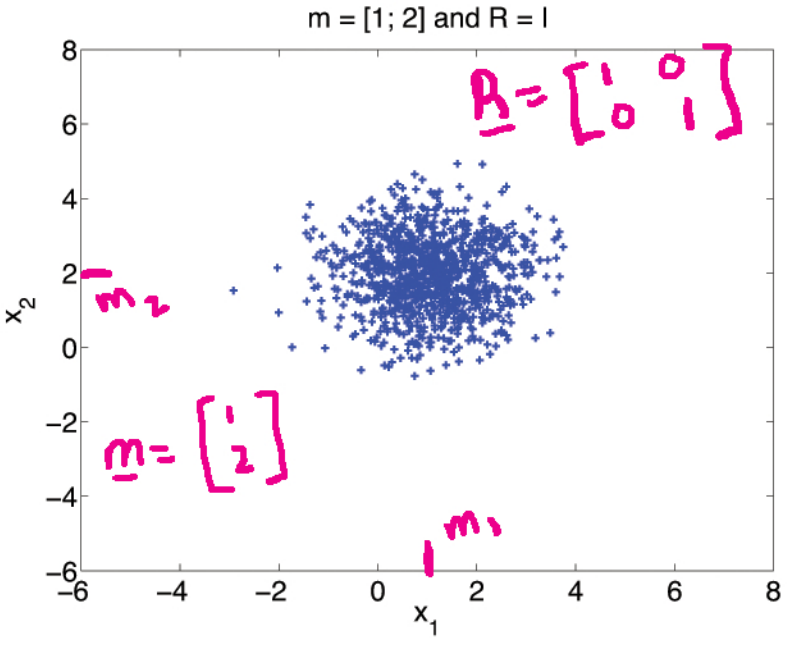
$r_{ij} = 0$ "on average" no consistent relation between $x_i - m_i$ and $x_j - m_j$ uncorrelated

As $|r_{ij}|$ increases, degree of coupling between x_i and x_j increases

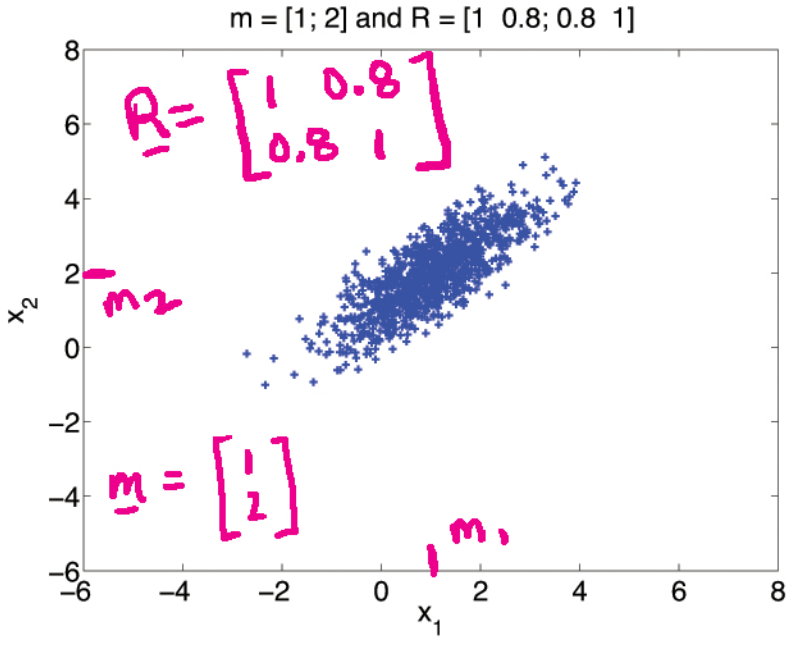
Correlation Coefficient $$\rho_{ij} = \frac{r_{ij}}{(r_{ii}\, r_{jj})^{1/2}} \qquad -1 < \rho_{ij} < 1$$

Note - $|r_{ij}| \leq (r_{ii} r_{jj})^{1/2}$

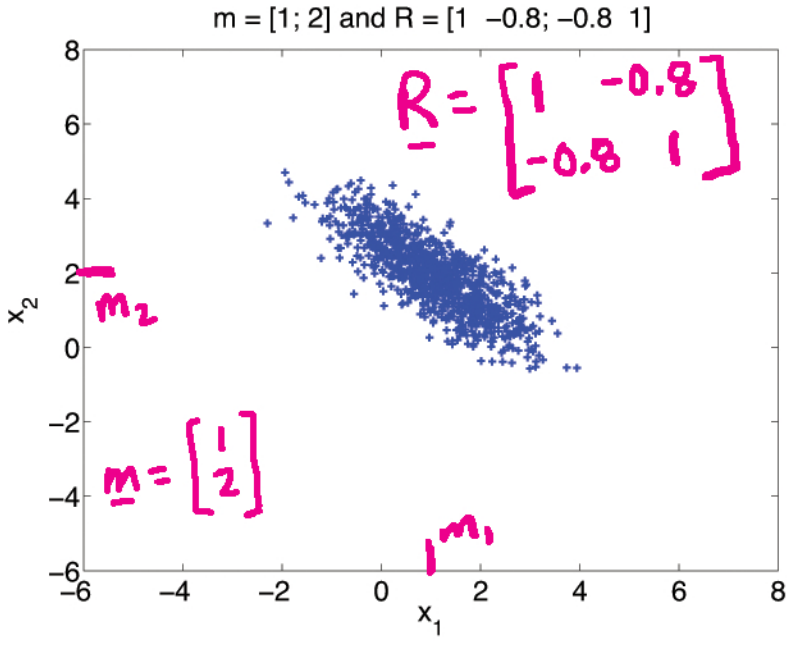
$r_{12} = 0$



$r_{12} > 0$



$r_{12} < 0$



Multivariate Gaussian Geometry

$f(\underline{x}) = \text{const}$ (level curves) $\Rightarrow$ $(\underline{x}-\underline{m})^T \underline{R}^{-1} (\underline{x}-\underline{m}) = c$

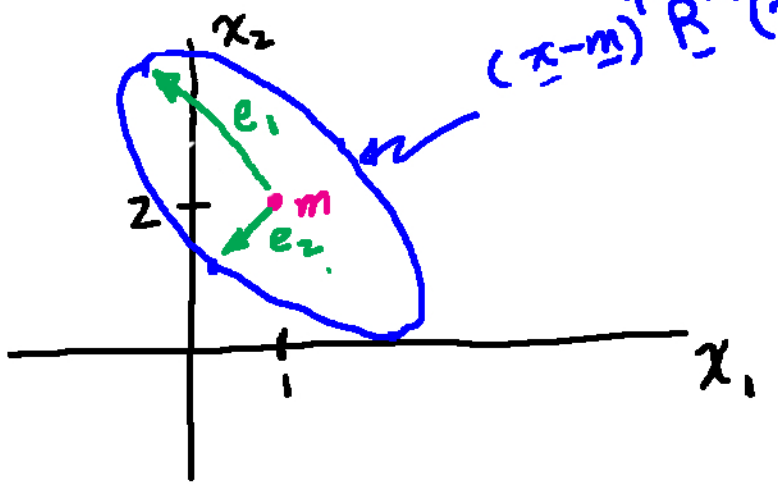
ellipse ($N=2$)
ellipsoid ($N>2$)



Centered at $\underline{x} = \underline{m}$

Major axis: direction of eigenvector for largest eigenvalue of $\underline{R}$
length $2\sqrt{c\lambda_{max}}$

Minor axis: direction of eigenvector for smallest eigenvalue
length $2\sqrt{c\lambda_{min}}$



Example: $\underline{m} = [1 \quad 2]^T$

$$\underline{R} = \begin{bmatrix} 1 & -0.8 \\ -0.8 & 1 \end{bmatrix}$$

$\lambda_1 = 1.8$, $\underline{e}_1 = \frac{\sqrt{2}}{2}\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$\lambda_2 = 0.2$, $\underline{e}_2 = \frac{\sqrt{2}}{2}\begin{bmatrix} -1 \\ -1 \end{bmatrix}$

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